START SMARTER

How an LLM Works

If you've used AI, you've probably used an app like ChatGPT, Claude, or Gemini. The app is what you use. Under the hood, a **Large Language Model**, or **LLM** for short, does the work.

What's an LLM?

Large

Trained on huge amounts of text and code.

Language

Reads, writes, summarizes, translates, and explains.

Model

Predicts likely output from learned patterns.

ChatGPT is the app. The LLM is the engine.

So how does an LLM actually turn your words into an answer?

When ChatGPT or Claude write a sentence, they're running math to predict the likely next words. They aren't looking up what your words mean; they're working out which words tend to follow which.

How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.

Learn Once. Answer Every Word.

LEARN ONCE

01 **Training**
The model learns from enormous amounts of data once, before you use it.

02 **Patterns**
Training turns examples into learned numerical patterns.

→ Patterns power every answer

ANSWER · EVERY WORD

03 **Probability**
For every next word, the model scores what is most likely.

04 **Prediction**
It chooses one likely next word, then runs the process again.

Learn once. Use the patterns for every answer.

Four ideas carry it. Below, we follow one example, peanut butter, through all four.

01 Training

The model **teaches itself**: guess the next word, check, and nudge its numbers toward the right word.

How Training Works

1 **Read**
The model reads a training example with the answer included.

2 **Guess**
Given 'Peanut butter and ___,' the model guesses cloud.

3 **Check**
The example says jelly. Compare the guess with that word.

4 **Adjust**
Adjust the model's internal numbers to make jelly more likely in this situation.

NEXT EXAMPLE

Repeat with more examples. The patterns build.

02 Patterns

So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?

How AI Learns Patterns

One Familiar Pattern

Peanut butter and ___ → jelly

You knew it. So does AI.

Patterns Are Everywhere

Twinkle, twinkle, little ___ → star

Once upon a ___ → time

Better late than ___ → never

AI learns patterns by working through billions of examples.

These are easy patterns you already know. AI also learns patterns in places you might not expect: how people explain ideas, ask questions, solve problems, and even misspell words.

03 Probability

AI doesn't make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.

Same Word. Different Odds.

I'd like to buy peanut butter and _____.

jelly	41%
bread	27%
bananas	16%
honey	5%

I'd like to buy a peanut butter and banana _____.

sandwich	54%
smoothie	16%
toast	9%
jelly	2%

The surrounding words change the odds.

In this example, adding "banana" drops the probability of "jelly" from 41% to 2%.

04 Prediction

Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next.

One Word at a Time

I want to buy peanut butter and
→ jelly

→

I want to buy peanut butter and jelly
→ for

→

I want to buy peanut butter and jelly for
→ lunch

Add a word. Use the updated sentence. Predict again.

A full paragraph runs this loop many times, fast enough to look like thought.

Training builds the patterns.

The model uses those patterns to build your answer, one word at a time.